

CSE 413 Optimization Theory and Applications

COURSE SYLLABUS · [Fall 2026 – 2027]

Instructor	Dr. Taha Yiğit Alkan	Office Hours	Artificial Intelligence and Data Science Application & Research Center Wednesday, 13:30-15:30
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Course Description

This course teaches how to model engineering problems as mathematical optimization problems and how to select and apply the solution method appropriate to the problem's structure. Linear programming and duality, gradient-based and Newton-type methods, KKT conditions, convex optimization, stochastic gradient methods, integer programming, dynamic programming and network optimization, and metaheuristics are covered with a balance of theory and practice.

Learning Outcomes

Upon successful completion of this course, the student will be able to:

1. Model engineering problems as mathematical optimization problems.
2. Explain the fundamental optimization algorithms with their theoretical foundations.
3. Certify the optimality of a solution using duality and KKT conditions and perform sensitivity analysis.
4. Apply optimization techniques to computer engineering problems.

Assessment

Component	Weight	Scope and Dates
Midterm Exam	30%	Week 8. Scope: topics of Weeks 1–7. One handwritten A4 formula sheet is allowed.
Project	30%	Students are expected to present their final work.
Final Exam	40%	Final exam week. Scope: Weeks 1–14 (all topics). One handwritten A4 formula sheet is allowed.

Weekly Schedule

Week	Topics	Laboratory / Deliverables
1	Introduction to optimization: modeling, decision variables, objectives, constraints	Python/Jupyter environment; surfaces, contour plots, a tour of scipy.optimize
2	Mathematical foundations: norms, gradients, Taylor expansion, a first look at convexity	Gradient checking, quadratic forms, numerical computing exercises
3	Linear programming I: geometry and the simplex method	Tableau simplex from scratch; bandwidth-allocation case study
4	Linear programming II: duality, shadow prices, sensitivity analysis	Duals with PuLP/CBC; RHS/objective ranging

5	Unconstrained optimization I: gradient descent, step-size selection, convergence rates	Implementing GD, Armijo/backtracking; sensor-calibration case
6	Unconstrained optimization II: Newton, quasi-Newton (BFGS, L-BFGS), Gauss–Newton	Method race; a tour of <code>scipy.optimize</code>
7	Lagrange multipliers and KKT conditions	A KKT checker; water-filling; barrier interior-point method
8	Convex optimization I: sets, functions, the problem hierarchy (LP–QP–SOCP–SDP)	Convexity tests; introduction to CVXPY
9	Convex optimization II: modeling with CVXPY, DCP, dual variables	Application gallery: SVM, Markowitz, Lasso
10	Stochastic gradient methods: SGD, momentum, Adam, proximal methods	SGD/momentum/Adam from scratch; schedules; ISTA
11	Optimization in machine learning and deep learning: backpropagation, automatic differentiation	micrograd; NumPy MLP vs. PyTorch face-off
12	Integer programming: modeling, LP relaxation, branch and bound, cutting planes	Branch and bound from scratch; sudoku as an ILP; facility location
13	Dynamic programming and network optimization: Bellman, Dijkstra, A*, max flow	Routing case study (national backbone); value iteration
14	Metaheuristics: simulated annealing, GA, PSO, artificial bee colony; VLSI applications	SA/GA/PSO/ABC shootout; floorplanning + gate sizing
15	Term project presentations (15 min + live demo)	—

Course Materials

Primary materials: the instructor's weekly lecture decks and hands-on Jupyter lab notebooks (shared on the course page). Supplementary resources:

- Boyd, S., & Vandenberghe, L. (2004). *Convex optimization*. Cambridge University Press. (free e-book)
- Nocedal, J., & Wright, S. J. (2006). *Numerical optimization* (2nd ed.). Springer.
- Winston, W. L. (2004). *Operations research: Applications and algorithms* (4th ed.). Brooks/Cole.
- Taha, H. A. (2018). *Yöneylem araştırması [Operations research]* (Ş. A. Baray & Ş. Esnaf, Trans.). Literatür.
- Pınar, M. Ç. (2020). *Dışbükeylik ve optimizasyon: Ders notları [Convexity and optimization: Lecture notes]*. Bilkent University.
- Karaboğa, D. (2014). *Yapay zeka optimizasyon algoritmaları [Artificial intelligence optimization algorithms]*. Nobel.

Course Policies

Academic integrity: Discussing ideas is encouraged. Submitted code and reports must be individual work (team work within the project team). Code plagiarism, cheating, and undisclosed sources are grounds for disciplinary action.

AI tools: Use for learning is welcome. If AI tools are used in assignments or the project, how they were used must be declared in the report. All AI tools are prohibited in exams. Undeclared use is treated as plagiarism.

Communication: Please use the course discussion board for questions first. MS Teams messages are answered within 3 business days.